

Mini-Project: Freezer Optimization

ECE 271C – Dynamic Programming

May 4, 2026

1 Project Overview

This one-week mini-project explores the application of dynamic programming and optimization techniques to industrial freezer operation under varying electricity pricing conditions. You will develop a cost-minimizing control strategy for freezers while maintaining temperature constraints.

2 Background: Industrial Freezer Operation

Industrial freezers offer unique opportunities for energy cost management due to their thermal mass. Unlike many electrical loads, freezers can “store cold” – i.e., they can be pre-cooled during periods of low electricity prices and then allowed to warm (within limits) during high-price periods.

Key characteristics:

- Freezers have thermal inertia that creates flexibility in when cooling is applied
- Electricity costs vary throughout the day (time-of-use pricing)
- Demand charges (fees based on peak power consumption) may apply
- Operating constraints include temperature limits for food safety and equipment protection

3 Freezer Thermal Model with Uncertainty

The temperature dynamics of an industrial freezer can be modeled using a discrete-time difference equation that captures both deterministic physics and stochastic disturbances. The evolution of freezer temperature from one time step to the next is modeled as:

$$T_{k+1} = T_k + \alpha(T_{\text{amb}} - T_k) - \beta P_k + w_k \quad (1)$$

where:

- T_k is the freezer temperature at time step k [C]
- T_{amb} is the ambient temperature [C]
- P_k is the cooling power at time step k [kW]
- α is a thermal leakage parameter (typically 0.05 to 0.15 per hour)
- β is a cooling efficiency parameter (typically 0.3 to 0.7 C/kWh)

Hour	Time	Ambient Temperature (C)		
		Hot Desert (Phoenix, AZ)	Temperate Coastal (San Francisco, CA)	Cold Northern (Minneapolis, MN)
0	12:00 AM	27.5	14.8	5.2
1	1:00 AM	26.3	14.5	4.6
2	2:00 AM	25.2	14.2	4.0
3	3:00 AM	24.5	13.9	3.5
4	4:00 AM	23.8	13.7	3.0
5	5:00 AM	23.5	13.5	2.8
6	6:00 AM	24.2	13.7	3.2
7	7:00 AM	26.0	14.2	4.5
8	8:00 AM	28.4	15.0	6.3
9	9:00 AM	31.2	16.1	8.7
10	10:00 AM	33.8	17.3	11.2
11	11:00 AM	35.9	18.2	13.5
12	12:00 PM	37.5	19.0	15.3
13	1:00 PM	38.7	19.6	16.8
14	2:00 PM	39.4	20.0	17.5
15	3:00 PM	39.8	20.2	17.8
16	4:00 PM	39.5	19.8	17.2
17	5:00 PM	38.2	19.0	16.0
18	6:00 PM	36.5	18.1	14.3
19	7:00 PM	34.3	17.0	12.5
20	8:00 PM	32.1	16.2	10.2
21	9:00 PM	30.5	15.7	8.4
22	10:00 PM	29.2	15.3	7.0
23	11:00 PM	28.3	15.0	6.0

Table 1: Hourly ambient temperature profiles for three different geographical locations. These temperature profiles can be used to analyze how geographical location affects optimal freezer operation strategies.

- w_k is the uncertainty term representing stochastic disturbances [C]
- $\alpha(T_{\text{amb}} - T_k)$: Heat gain from the environment. This term models the natural heat transfer from the warmer ambient environment to the colder freezer interior.
- βP_k : Active cooling effect. This term represents the temperature reduction achieved by applying cooling power P_k .
- w_k : Uncertainty component. This term captures all unpredictable thermal disturbances that affect the freezer temperature, primarily caused by product loading/unloading and door openings.

Table 1 provides an average ambient temperature analysis for three different facilities located in different geographical regions.

3.1 Modeling the Uncertainty

For this project, you should model w_k as a random variable with the following characteristics:

- Mean: $\mu_w = 0.5$ C (slight positive bias representing typical operations)
- Standard deviation: $\sigma_w = 1.0$ C
- Distribution: You may assume Gaussian, but feel free to explore other distributions if justified

A more realistic model of uncertainty would be in the form of a time-varying uncertainty model as given in the following table.

Time Period	Expected Disturbance μ_k (C)	Standard Deviation σ_k (C)	Business Activity
00:00-05:59	0.0	0.3	Minimal (night)
06:00-07:59	0.5	0.5	Morning deliveries
08:00-09:59	2.0	1.0	Major morning loading
10:00-11:59	1.0	0.8	Moderate activity
12:00-13:59	0.5	0.5	Lunch break activity
14:00-15:59	1.5	0.8	Afternoon loading
16:00-17:59	2.0	1.0	End-of-day loading
18:00-23:59	0.2	0.4	Evening (reduced activity)

Table 2: Uncertainty profile by time of day

4 Electricity Cost Components

The total cost to be minimized is:

$$C_{\text{total}} = C_{\text{energy}} + C_{\text{demand}}. \quad (2)$$

The energy cost over a planning horizon is given by

$$C_{\text{energy}} = \sum_{k=0}^{N-1} c_k \cdot P_k \cdot \Delta t \quad (3)$$

where:

- c_k is the electricity price at time step k [\$/kWh]
- P_k is the power consumption at time step k [kW]
- Δt is the duration of each time step [h]
- N is the number of time steps in the planning horizon

For this project, use the following time-of-use pricing structure:

Many industrial electricity tariffs also include demand charges based on peak consumption:

$$C_{\text{demand}} = c_d \cdot \max_{k \in \{0,1,\dots,N-1\}} P_k \quad (4)$$

Time Period	Price (USD/kWh)
00:00 - 05:59	0.05
06:00 - 07:59	0.10
08:00 - 17:59	0.20
18:00 - 19:59	0.10
20:00 - 23:59	0.05

Table 3: Time-of-use electricity pricing

where c_d is the demand charge rate [\$/kW]. A typical value associated with the demand charge rate is \$18.50/kW.

This demand charge is calculated over a 30-day billing period. For simplicity in this project, assume that the optimal control strategy you develop for a 24-hour cycle would be repeated each day of the month, so your maximum power consumption in the 24-hour period would determine the monthly demand charge.

5 Operational Constraints

- Temperature bounds: $T_{\min} \leq T_k \leq T_{\max}$ (typically -25°C to -15°C)
- Maximum cooling power: $0 \leq P_k \leq P_{\max}$
- Initial temperature: $T_0 = -20^{\circ}\text{C}$ (unless specified otherwise)
- Final temperature: $T_N \leq T_{\max}$ (optional constraint)

6 Electricity Cost Components

6.1 Time-Varying Electricity Pricing

The energy cost over a planning horizon is:

$$C_{\text{energy}} = \sum_{k=0}^{N-1} c_k \cdot P_k \cdot \Delta t \quad (5)$$

where:

- c_k is the electricity price at time step k [\$/kWh]
- P_k is the power consumption at time step k [kW]
- Δt is the duration of each time step [h]
- N is the number of time steps in the planning horizon

6.2 Demand Charges

Many industrial electricity tariffs include demand charges based on peak consumption:

$$C_{\text{demand}} = c_d \cdot \max_{k \in \{0,1,\dots,N-1\}} P_k \quad (6)$$

where c_d is the demand charge rate [\$/kW]. This demand charge is calculated over a 30 day billing period.

6.3 Total Cost

The total cost to be minimized is:

$$C_{\text{total}} = C_{\text{energy}} + C_{\text{demand}} \quad (7)$$

7 Operational Constraints

- Temperature bounds: $T_{\min} \leq T_k \leq T_{\max}$ (typically -25C to -15C)
- Maximum cooling power: $0 \leq P_k \leq P_{\max}$
- Initial temperature: T_0 (given)
- Final temperature: $T_N \leq T_{\max}$ (optional constraint)

8 Project Goals and Scenarios

This project challenges you to apply dynamic programming techniques to industrial freezer optimization. You'll work through increasingly complex scenarios that build upon each other, culminating in an equipment selection analysis.

8.1 Core Learning Objectives

- Apply dynamic programming principles to a real-world industrial scenario
- Model thermal dynamics with stochastic components
- Formulate and solve multi-stage decision problems under uncertainty
- Evaluate trade-offs between energy costs, demand charges, and temperature constraints
- Analyze capital investment decisions using optimization results

8.2 Project Scenarios

You will work through three scenarios of increasing complexity:

1. Scenario 1: Basic Freezer Optimization with Time-Varying Pricing

- Implement the freezer thermal dynamics model
- Create a dynamic programming solution for a 24-hour horizon
- Find the optimal cooling strategy to minimize electricity costs
- Analyze how the freezer “stores cold” during low-price periods
- Compare your optimized strategy with a simple thermostat control

2. Scenario 2: Adding Demand Charges and Geographical Variations

- Incorporate demand charges into your cost function
- Analyze how demand charges affect the optimal control strategy

- Evaluate how freezer operation should adapt to different climate zones
- Identify critical periods and constraints that drive costs
- Quantify potential savings compared to simpler control strategies

3. Scenario 3: Compressor Selection and Capital Investment Analysis

- Evaluate three different compressor options across multiple locations
- Calculate annual operating costs using your optimized control strategies
- Perform ROI analysis considering capital costs, efficiency, and maintenance
- Determine which compressor provides the best lifecycle cost for each location
- Analyze sensitivity to key parameters (electricity prices, climate, usage patterns)

8.3 Baseline Comparison: Simple Thermostat Control

The simple thermostat control operates on a binary logic: if the freezer temperature rises above $T_{\text{high}} = -17\text{C}$, the compressor runs at maximum power P_{max} ; if temperature falls below $T_{\text{low}} = -23\text{C}$, the compressor turns off completely; otherwise, it maintains its previous state to prevent rapid cycling.

This non-optimized strategy serves as a baseline for comparing your dynamic programming approach. Unlike the optimized strategy, it does not consider electricity prices, demand charges, or forecasted conditions, and therefore cannot pre-cool during low-price periods or spread load to reduce peak demand.

8.4 Compressor Options

Characteristic	Compressor A	Compressor B	Compressor C
Capital Cost	\$8,500	\$12,000	\$16,500
Maximum Power (P_{max})	8 kW	10 kW	12 kW
Cooling Efficiency (β)	0.35 C/kWh	0.5 C/kWh	0.65 C/kWh
Expected Lifetime	8 years	10 years	12 years
Maintenance Cost/Year	\$600	\$450	\$350

Table 4: Comparison of available compressor options for the freezer system

8.5 Deliverables

For each scenario, you should provide:

1. Problem Formulation:

- Mathematical definition of state space, action space, and transition dynamics
- Clear statement of objective function and constraints
- Description of your dynamic programming approach

2. Numerical Results:

- Optimal control policies
- Cost breakdown (energy vs. demand charges)
- Temperature and power consumption profiles
- Comparative analysis with baseline strategies

3. **Business Insights:**

- Key factors driving operational costs
- Recommended control strategies for different conditions
- For Scenario 3: Compressor recommendation with justification
- Sensitivity analysis highlighting important design considerations

8.6 **Final Presentation**

Prepare a 30-minute presentation that includes:

- Problem formulation and solution approach
- Key results and visualizations from all three scenarios
- Insights on optimal freezer operation across different conditions
- Compressor selection recommendations with ROI analysis
- Challenges encountered and how you addressed them

8.7 **Evaluation Criteria**

Your project will be evaluated based on:

- Correctness of problem formulation and dynamic programming implementation
- Appropriateness of solution method and numerical analysis
- Quality of insights and business recommendations
- Depth of sensitivity analysis and consideration of real-world factors
- Clarity and organization of final presentation